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PI Name:	Behringer, Robert P. Ph.D.		
Project Title:	Gravity and Granular Materials		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bob@phy.duke.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	919 660-2550
Organization Name:	Duke University		
PI Address 1:	Department of Physics		
PI Address 2:	Physics Bldg, Campus Box 90305		
PI Web Page:	http://behringer.phy.duke.edu/		
City:	Durham	**State:**	NC
Zip Code:	27708-0305	**Congressional District:**	1
Comments:	NOTE: PI Robert Behringer is deceased, as of July 2018.		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
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Contact Monitor:	Wilkinson, R Allen	**Contact Phone:**	216-433-2075
Contact Email:	R.A.Wilkinson@grc.nasa.gov		
Flight Program:			
Flight Assignment:	Note- grant renewed for 4 years/new NAG#		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kondic, Lou (New Jersey Institute of Technology)		
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	This project entails the study of the properties of granular materials. The presence of gravity plays a strong role in how granular materials behave. In some instances, it also obscures fundamental behavior that could be much better understood in the presence of reduced gravity. The objectives of this work are to 1) understand the role that gravity plays in certain granular systems, such as sheared and shaken systems, and 2) to uncover basic science that governs the way granular materials behave and that is obscured ordinarily by the presence of gravity, 3) from the increased understanding from 1 and 2, increase our ability to predict and control granular flows, thus leading to significant savings in technical and commercial processing industries in the US and abroad, and 4) provide insight into granular materials handling in reduced-gravity situations. 5) In recognition of the initiative to explore the Moon and Mars, we particularly focused on a) improving our understanding of predictability for granular materials, b) understanding the scope of granular processing for Lunar and Maritan exploration, and c) investigating novel systems of 2-phase granular-gas flow		

Task Description:	that may be highly important for granular handling in the above environments. The initial task consisted of a detailed characterization of the dynamics of shaken granular materials. Experiments identified important behavior and flow properties for shaken granular materials. Shaking was carried out both vertically and horizontally. The vertical component of shaking provided a means to modify gravity, and the horizontal component provided oscillating shear stress. This technique provided an excellent method to explore concepts of granular friction and failure. Next. the task focused on the properties of slowly sheared dense granular materials. Using a novel apparatus that allows characterization of all aspects of this process in 2 dimensions, we have discovered a novel transition that likely plays a key role in the failure of granular materials under applied stress. This is a very complex system, and additional work focused on understanding its properties. These explorations was extended to a much larger 2D system. This system was used to probe the nature of diffusive, affine and non-affine motion in a physical granular system. The task was extend to include a flight mission using a 3D system. In this regard, a key point is that on earth gravity masks the true dynamical properties of both rapidly and slowly sheared granular systems. At this stage, however, the flight aspect of this project is no longer active, due to changes in NASA goals and missions. Significant effort went into the design of the project, the development of an experiment at Duke University, development of mission plans, and the development of a breadboard device. The study of 3D systems lead to the development of a highly novel Couette shear apparatus that allows us to simultaneously shear a granular sample (from above) while providing energy from below by vibration. We have used this apparatus to probe the jamming/unjamming transition in dense granular materials. This apparatus led to the discovery of a fascinating transition from a crystallized state to a disordered state in this system, where order is induced by vibration and disorder is induced by shear. This transition is particularly striking because vibration, which is ordinarily associated with thermal disordering, is actually drives the ordering process. Here, it is important to note that gravity is present and likely playing an important role in these earthbound experiments. This experiment formed a basis for the development of a model for the development of a flight experiment. An additional part of the project involved detailed MD/discrete element studies of an experiment comparable to the 3D shear/shake experiment and the originally planned flight experiment. This involved work by Prof. Lou Kondic of NJIT to understand what we might expect from a flight experiment, and to address fundamental physics issues that arise in this type of system. Were NASA to return to ISS flights, this experiment would continue to provide an important opportunity for understanding granular mechanics/statistics. A initial version of this experiment was built and used at Duke. A second version was built under NASA auspices, by ZIN-TECH. This latter version of the experiment has been put to use in the lab of Prof. Karen Daniels, now at NC State University. One final aspect has been the characterization of meteor-like impacts through the novel use of photoelastic materials. This study has the potential to inform the energetics of meteor impacts on surfaces such as the moon.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A deeper understanding of the properties of granular materials is essential for a broad spectrum of earth-bound tasks. Failures of devices on the Earth are all too common, and much insight is needed. The present tasks should help significantly towards this end. The shaking and shearing of granular materials are particularly important as basic means to cause flow in granular materials, and hence as tools to transport and process a variety of materials, including coal, ores, food grains, pharmaceutical powders and even dog food. These processes also plays key roles in natural phenomena such as avalanches and earthquakes. Gravity plays an intrinsically important role in these cases. Looking into the future, attempts to mine on the lunar surface will almost surely involve shaking processes for granular materials. Because our understanding of granular flows is rather poor, many of the plants and devices which handle granular materials operate significantly below capacity, and fail all too frequently. The result is corresponding loss to the US economy. The expectation is that increased scientific insight obtained from this project will lead to improved predictability and efficiency in a range of technological processes.
	We have developed and used a highly novel Couette shear apparatus that allows us to simultaneously shear a granular sample (from above) while providing energy from below by vibration. We have used this apparatus to probe the jamming/unjamming transition in dense granular materials. We find a fascinating transition from a crystallized state to a disordered state in this system, where order is induced by vibration and disorder is induced by shear. There is a phase boundary between ordered and disordered states that corresponds to equal energy input per particle from shear and vibration. This transition is particularly striking because vibration, which is ordinarily associated with thermal disordering, is actually drives the ordering process. Here, it is important to note that gravity is present and likely playing an important role in these earthbound experiments. We expect that work will continue on this project to further understand the competing roles of energy input, and other properties of this system. This experiment also formed a basis for the development of a model for the development of a flight experiment. In addition, we are worked closely with Prof. Lou Kondic of NJIT to understand what we might expect from a flight experiment, and to address fundamental physics issues that arise in this type of system. Were NASA to return to ISS flights, this experiment would continue to provide an important opportunity for understanding granular mechanics/statistics. A initial version of this experiment was built and used at Duke. A second version was built under NASA auspices, by ZIN-TECH. This latter version of the experiment has been put to use in the lab of Prof. Karen Daniels, now at NC State University. We have used this apparatus to probe the jamming/unjamming transition in dense granular materials. We find a fascinating transition from a crystallized state to a disordered state in this system, where order is induced by vibration and disorder is induced by shear. There is a phase boundary between ordered and disordered states that corresponds to equal energy input per particle from shear and vibration. This transition is particularly striking because vibration, which is ordinarily associated with thermal disordering, is actually drives the ordering process. Here, it is important to note that gravity is present and likely playing an important role in these earthbound experiments. We expect that work will continue on this project to further understand the competing roles of energy input, and other properties of this system. This experiment also formed a basis for the development of a model for the development of a flight experiment. In addition, we are worked closely with

Task Progress:	Other experiments used 2D Couette flow, which effectively eliminates gravity, to probe the nature of diffusion, affine and non-affine motion. The apparatus used here was a significantly larger version of a type of apparatus that we used to discover a novel jamming transition in shear granular material. In the earliest aspect of this project, we probed the nature of friction, gravity and vibration using novel experiments that allowed simultaneous vertical and horizontal vibration. Here, we have carried out several experimental projects, including tests for alternative variables or a granular temperature. First, in the context of Couette shear in a very large 2D system consisting of photoelastic particles, we measured the self-diffusion of particles. It appears that diffusion is Brownian, although actual trajectories are strongly influenced by gradients in the mean flow. We have carried out {\bf experimental tests of mobility and diffusion}. Motivation for such studies included the predictions by Makse and Kurchan, experimental studies by Albert et al., and the predictions by Radjai and Roux already noted. The first of our experiments yielded diffusivity, D, and mobility, B, for a stirred granular system. Although we observed simple Brownian diffusion in this system, we did not obtain sensible results for the mobility. Rather, the force needed to push the tracer particle through the other particles was nearly independent of velocity, showing only a logarithmic increase over a broad range. This behavior resembles Coulomb friction more than a molecular fluid, and precludes defining a temperature in terms of the ratio of D/B for this arrangement. We believe that the reason for the discrepancy between the Makse simulations and our experiments lies in the fact that for constant (moderate) speed pushing, the tracer breaks force chains rather than simply taking advantage of random `thermal' fluctuations that provide opportunities for tracer motion.  In a rather different set of experiments, we are beginning to obtain results testing the idea of Edwards entropy or compactivity as relevant `thermodynamic' variables using a novel apparatus that consists of two different size spheres, lying on a plane and subject to shaking. There is a movable barrier between the two particle types, and as the system is shaken, the position of the barrier fluctuates/evolves in time. Regarding rate dependence, we carried out extensive 2D Couette shear experiments, where we determined the force on particles using photoelasticity. For at least five decades in shear rate, the mean force per particle (roughly the pressure) grows logarithmically with shear rate. This rate dependence is presumably related to that seen in sophisticated measurements of friction, and in rate-and-state models. We extensively studied force transmission of a static system in response to a point load in 2D using photoelastic particles. These studies showed two-peaked wave-like transmission for geometrically ordered packings, with less broadening of the peaks as friction was decreased. For disordered systems of moderate size, the response was elastic-like. We carried out extensive studies of how granular materials interact frictionally with boundaries, again using 2D photoelastic techniques. In one of these experiments, particles were confined to a 2D version of a silo. We determined the response of this system to an applied load from above, to a driving force from below, etc. Unlike past studies of model silos, we were able to determine forces internally, and investigate the uniformity of stresses across a horizontal slice. An important aspect of our photoelastic studies, carried out by Ph.D.~student Trushant Majmudar, has been to develop algorithms and computer code to extract vector forces at contacts between grains. In previous work, we used an empirical calibration relating the gradients of intensity in a photoelastic image to the force on a particle. The methods that we now have implemented can automatically process an image, compute the stress field within each particle, and determine the vector forces at contacts. A recent novel topic is impacts of objects such as meteors on soils. This issue is also quite relevant to Lunar exploration. Real meteors are massive and release large amounts of energy on impact, but some essential aspects of the physics can be probed in controlled experiments. A number of novel studies have considered the dynamics of lab-scale impact events. We have explored the effect of impacts using 2D particles and high-speed video as a novel technique to directly watch energy flow during the impact of a large intruding particle. Much of this work has been carried out by Joyce Coppock, as undergraduate student at Duke. This work has appeared in the Journal Chaos. Joyce is currently enrolled as a Ph.D. student at the University of Maryland. We have carried out several numerical modeling projects, two of which were in collaboration with Prof. Lou Kondic of the New Jersey Institute of Technology. One involved generalizing the idea of granular temperature to flowing dense granular materials for large pressures and very slow speeds, a regime in which the ordinary granular temperature is an imperfect measure of energy fluctuations. We proposed an extension of granular temperature drawn from the idea of equipartition:. We then compared this new temperature to a granular temperature drawn from energy fluctuations, and found that these two temperatures were in reasonable agreement for our model system, whereas as for the given conditions, the standard granular temperature, was much too small to be representative of energy fluctuations. We are continuing work in this direction with the added collaboration of Prof. Corey O'Hern of Yale University. The computations for the purpose of understanding velocity profiles and stresses in sheared systems were carried out using two different approaches: (i) previously developed soft-sphere model where dynamics of the particles is calculated by computing forces that particles experience at each time, and (ii) hard-sphere model where one finds the time of the next collision of a given particle, evolves the system and then updates appropriate variables as a collision is taking place. The soft-sphere simulations were mostly carried out in two space dimensions (2D), while the hard-sphere simulations were carried out in three dimensions (3D). Formulating and carrying out these 3D simulations was the main project of the postdoctoral associate, Oleh Baran, who has developed an approach for efficiently computing the dynamics in the domain closely resembling the experimental one. The outcome of these simulations were precise predictions of the particles' dynamics in various configurations: with and without gravitational effects, for different physical properties of the boundaries, with and without additional oscillatory driving, to name a few. The simulations made clear predictions regarding the influence of gravity on the velocity profiles in addition to quantifying the effects of the wall friction and related effects. Furthermore, we identified non-dimensional parameters involving the effects of gravity, shear, and oscillatory driving governing the system behavior.
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